

Simplified Notes on the Open Problem: Can Local Regularization Learn All Multiclass Problems?

1 Simplified Statement of the Problem

We are in a multiclass classification setting.

Basic objects

- X : input/domain space
- Y : label space
- $\mathcal{H} \subseteq Y^X$: hypothesis class
- D : realizable distribution over $X \times Y$
- $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$: i.i.d. training sample from D

A hypothesis $h \in Y^X$ maps inputs $x \in X$ to labels $h(x) \in Y$.

Loss function

The classification loss is the 0-1 loss:

$$\ell_{0-1}(h(x), y) = \begin{cases} 0, & h(x) = y, \\ 1, & h(x) \neq y. \end{cases}$$

True risk

The true error of h under distribution D is

$$L_D(h) = \mathbb{E}_{(x,y) \sim D} [\ell_{0-1}(h(x), y)].$$

Empirical risk

The empirical error on the sample S is

$$L_S(h) = \frac{1}{n} \sum_{i=1}^n \ell_{0-1}(h(x_i), y_i).$$

2 What Are We Solving For?

The learner receives a sample S and must output a predictor $A(S)$ such that

$$L_D(A(S)) \leq \varepsilon$$

with probability at least $1 - \delta$, for sufficiently many samples.

So the unknown being “solved for” is:

$\text{a learning rule } A \text{ (or equivalently a predictor } A(S))$

with low true error.

In this paper, the specific question is whether we can solve the multiclass learning problem using a *local regularizer*.

3 The Main Optimization Form

In binary learning, ERM chooses a hypothesis with minimum empirical error:

$$\operatorname{argmin}_{h \in \mathcal{H}} L_S(h).$$

For multiclass learning, the paper studies *local regularization*.

A local regularizer is a function

$$\psi : \mathcal{H} \times X \rightarrow \mathbb{R}_{\geq 0}.$$

For a test point x , the learner looks at hypotheses with zero empirical error and chooses a prediction from those minimizing the local penalty:

$$A(S)(x) \in \left\{ h(x) : h \in \operatorname{argmin}_{h \in L_S^{-1}(0)} \psi(h, x) \right\}.$$

So the mathematical object we are trying to find is:

$\psi(h, x)$

such that the induced learner PAC learns the hypothesis class.

4 Simplified Core Question

The paper’s main question can be rewritten as:

Given any learnable multiclass hypothesis class $\mathcal{H}$, does there always exist a local regularizer $\psi(h, x)$ such that the induced learner learns $\mathcal{H}$?

Equivalently:

$\text{Does every PAC-learnable multiclass class admit a good local regularizer?}$

5 Open Problem 1

Formal version

Can all learnable multiclass hypothesis classes be learned by a local regularizer? If yes, can this be done with optimal or nearly optimal sample complexity?

Simplified version

Is local regularization the multiclass analogue of ERM?

In binary classification:

finite VC dimension $\implies$ ERM works.

The paper asks whether in multiclass classification:

learnable $\implies$ local regularization works.

6 Candidate Counterexample: The Class $\mathcal{H}_\Delta$

The paper constructs a special hypothesis class $\mathcal{H}_\Delta$, which may be hard for local regularization.

Domain and label set

Let

$$X = \mathbb{N}, \quad Y = \{*\} \cup 2^X.$$

So labels are either:

- the default symbol $*$, or
- a subset of X .

Triples of sets

Consider triples (A, B, C) such that:

$$|A| = |B| = |C| =: k < \infty,$$

and

$$|A \cap B| = |A \cap C| = |B \cap C| = \frac{k}{2},$$

with

$$A \cap B \cap C = \emptyset.$$

Hypotheses in $\mathcal{H}_\Delta$

For each such triple (A, B, C) , define hypotheses $h : X \rightarrow Y$ satisfying:

$$\begin{aligned} h(x) &= * && \text{for all } x \notin A \cup B \cup C, \\ h &\text{ is constant on } A \cap B, A \cap C, B \cap C. \end{aligned}$$

The values on the overlap regions are chosen as:

$$\begin{aligned} x \in A \cap B &\implies h(x) \in \{A, B\}, \\ x \in A \cap C &\implies h(x) \in \{A, C\}, \\ x \in B \cap C &\implies h(x) \in \{B, C\}. \end{aligned}$$

Since there are 2 choices on each of the 3 overlap regions, there are

$$2^3 = 8$$

such hypotheses for each triple (A, B, C) .

Then $\mathcal{H}_\Delta$ is the union of all such hypotheses over all valid triples.

7 Open Problem 2

Formal version

Can the class $\mathcal{H}_\Delta$ be learned by a local regularizer? If yes, with optimal or nearly optimal sample complexity?

Simplified version

Does $\mathcal{H}_\Delta$ defeat local regularization, or can local regularization still handle it?

Is $\mathcal{H}_\Delta$ a counterexample to Open Problem 1?

8 Why the $\mathcal{H}_\Delta$ Construction Is Tricky

The class $\mathcal{H}_\Delta$ is tricky for several reasons.

1. Many labels are themselves sets

The labels are not simple class names like 1, 2, 3, but actual subsets $A, B, C \subseteq X$. This makes the geometry of the problem much richer and more confusing.

2. The hypotheses look identical on much of the domain

Outside $A \cup B \cup C$, every hypothesis outputs *. So most of the domain gives no useful signal.

3. Ambiguity lives only on overlaps

The only interesting behavior is on:

$$A \cap B, \quad A \cap C, \quad B \cap C.$$

On each overlap, the hypothesis may choose either of two labels. So the important distinctions are hidden in small structured regions.

4. There is strong combinatorial symmetry

The sets A, B, C play nearly interchangeable roles. This makes it difficult for a learner to know which label is the “correct natural one” unless it has enough structural information from the sample.

5. The learner may need geometric information about the sample

A good predictor seems to need more than just:

- the test point x , and
- the empirical error values of hypotheses.

It may need to understand how the observed sample sits inside the hidden set structure.

9 Why Symmetry Breaks ERM

ERM chooses any hypothesis with minimum empirical error:

$$\operatorname{argmin}_{h \in \mathcal{H}} L_S(h).$$

The problem is that many hypotheses can fit the training sample equally well.

Key issue

Suppose the true labels come from set A , and the sample only shows some points of A . For a new test point $x_{\text{test}} \in A$ that did not appear in training, there may exist another hypothesis that:

- matches all training data perfectly,
- but labels x_{test} as B instead of A .

Thus both hypotheses have

$$L_S(h) = 0,$$

but they disagree on the test point.

Why ERM fails

ERM sees only empirical error, so it treats these hypotheses as equally good. Because of symmetry, it has no built-in reason to prefer A over some alternative B that also fits the sample.

So:

symmetry creates many zero-training-error hypotheses, and ERM cannot distinguish them

Hence ERM may generalize badly even though the class is learnable.

10 Why Local Regularization Might Also Fail

A local regularizer chooses among zero-error hypotheses using $\psi(h, x)$.

So instead of just minimizing $L_S(h)$, it minimizes:

$$\psi(h, x) \quad \text{over hypotheses with } L_S(h) = 0.$$

The concern is:

Can any function of only h and the test point x contain enough information to break the symmetry correctly?

The paper suspects the answer may be no for $\mathcal{H}_\Delta$.

Intuitively, a learner may need to inspect more of the sample geometry than local regularization allows.

11 How Researchers Might Try to Prove the Open Problem

There are two broad directions.

11.1 Approach A: Prove local regularization is universal

To prove Open Problem 1 positively, one would try to show:

For every learnable multiclass class $\mathcal{H}$, we can construct a local regularizer $\psi(h, x)$ that induces a PAC learner.

Possible strategy:

1. Start from an already-known optimal multiclass learner.
2. Show its predictions can be represented as minimizing some local score $\psi(h, x)$.
3. Prove the resulting learner has optimal or near-optimal sample complexity.

This would mean local regularization is expressive enough to simulate all learnable multiclass algorithms.

11.2 Approach B: Prove a counterexample

To prove Open Problem 1 negatively, one would try to show:

There exists a learnable multiclass class $\mathcal{H}$ such that no local regularizer can learn it.

For $\mathcal{H}_\Delta$, this would require proving that for every local regularizer ψ , there exists some realizable distribution D on which the induced learner fails.

Possible proof template:

1. Fix an arbitrary local regularizer $\psi(h, x)$.
2. Construct a hard realizable distribution D .
3. Use the symmetry of $\mathcal{H}_\Delta$ to show many hypotheses remain indistinguishable on the sample.
4. Show that minimizing $\psi(h, x)$ cannot always pick the correct label at test time.
5. Conclude that the learner has non-negligible error with noticeable probability.

12 The Main Mathematical Objects to Focus On

If you are studying this paper, the main variables and functions to track are:

X	: instance space,
Y	: label space,
$\mathcal{H}$	: hypothesis class,
D	: realizable distribution,
S	: sample,
h	: hypothesis,
$L_D(h)$	: true risk,
$L_S(h)$	: empirical risk,
$A(S)$	: learner output,
$\psi(h, x)$	: local regularizer.

The central unknown is:

$$\boxed{\psi(h, x)}$$

or else a proof that no such ψ can always work.

13 Final Simplified Summary

The paper is really asking:

In multiclass learning, is there one simple optimization-based principle that always works, the way ERM works in binary learning?

The candidate principle is local regularization:

$$A(S)(x) \in \left\{ h(x) : h \in \underset{h \in L_S^{-1}(0)}{\operatorname{argmin}} \psi(h, x) \right\}.$$

So the two key questions are:

1. Does every learnable multiclass class admit such a local regularizer?
2. If not, is $\mathcal{H}_\Delta$ the counterexample?